

Prudhveeraj Botta

Backend Software Engineer / Software Development Engineer (SDE) – Cloud, Distributed Systems

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SUMMARY

Software Engineer with **3+ years of experience** designing, building, and scaling distributed, cloud-native systems. Strong background in backend architecture, microservices, API security, performance engineering, and ML system integration. Proven track record of improving system reliability, reducing cloud costs, optimizing data pipelines, and accelerating CI/CD delivery. Experienced in production-grade engineering across AWS, large-scale databases, and AI/ML workflows. Adept at system design, cross-functional collaboration, and delivering high-impact solutions aligned with business and platform requirements.

Core Competencies: Backend Engineering, Cloud Architecture (AWS), Distributed Systems, API Design, Data Engineering, Security Engineering, ML Systems, DevOps & CI/CD

TECHNICAL SKILLS

- **Programming Languages:** Java, Python, JavaScript
- **Frameworks & Libraries:** Spring Boot, FastAPI, React.js, Next.js, TensorFlow
- **Backend & System Design:** Microservices, REST APIs, GraphQL, Event-Driven Architecture, API Gateway Patterns, High-Availability Systems
- **Distributed Systems:** Horizontal Scaling, Caching, Rate Limiting, Sharding, Consistent Hashing
- **Cloud & Infrastructure:** AWS (EC2, Lambda, ECS, Fargate, S3, DynamoDB, CloudWatch, IAM), Serverless Architecture, Docker
- **CI/CD & DevOps:** Jenkins, GitHub Actions, Automated Deployments, Infrastructure Optimization
- **Databases & Data Systems:** PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, Indexing, Data Modeling
- **Security & Reliability:** OAuth 2.0, JWT, RBAC, Encryption (AES), Secure API Design, Secrets Management, HIPAA Compliance
- **Testing & Performance:** JUnit, PyTest, Integration & Regression Testing, Load Testing, Profiling, Latency Optimization
- **ML Engineering:** TensorFlow Model Training & Deployment, ML Pipelines, Feature Engineering, Model Inference via APIs
- **Tools:** Git, Linux, Bash, Nginx, Postman

PROFESSIONAL EXPERIENCE

Dell Technologies – Remote, USA

Jan 2024 – Present

Software Developer

- Architected and scaled microservices-based backend systems using Java (Spring Boot), Python (FastAPI), and React.js, achieving 99.9% uptime across production workloads.
- Designed and deployed ML pipelines with TensorFlow for real-time inference, increasing data processing throughput by 35% and enabling predictive analytics at scale.
- Re-architected AWS infrastructure using Lambda, ECS, Fargate, auto-scaling, and IAM policy optimization, reducing cloud spend by 20% while maintaining performance SLAs.
- Built secure, production-grade APIs using OAuth 2.0, JWT, RBAC, and encryption, ensuring HIPAA compliance and reducing security incidents by 40%.
- Improved system performance with Redis caching, GraphQL query optimization, and rate limiting, cutting response latency by 50% and increasing throughput by 30%.
- Automated deployment pipelines with Docker, Jenkins, and GitHub Actions, reducing release cycles from weeks to days and increasing deployment reliability.
- Optimized PostgreSQL and DynamoDB schemas, indexing strategies, and access patterns, achieving 40% faster queries across distributed services.
- Collaborated with DevOps, ML, and security teams on architecture reviews; mentored engineers via design discussions and code reviews.

Bright Mind Technologies – India

Jul 2021 – Jul 2022

Software Developer

- Built full-stack applications using React.js (Next.js), Spring Boot, and Python (Flask/FastAPI), improving UI performance by 20% and backend service reliability.
- Deployed and scaled applications on AWS using load balancers, auto-scaling groups, and monitoring, maintaining 99.5% uptime.
- Implemented authentication, authorization, and RBAC using OAuth 2.0 and JWT, reducing unauthorized access by 35%.
- Developed automated testing frameworks (JUnit, PyTest, integration pipelines), improving regression coverage and increasing system efficiency by 25%.
- Engineered ETL and batch-processing pipelines, reducing processing time by 40% and enabling analytics-ready datasets.
- Participated in Agile planning, technical design, and peer reviews to ensure predictable delivery and long-term system maintainability.

Genius SoftTech – India

Sep 2020 – Jun 2021

Java Developer

- Developed scalable microservices using Java, Spring Boot, and Hibernate, improving modularity and reducing coupling in monolithic systems.
- Integrated ML models (TensorFlow, Scikit-learn) into backend services for predictive analytics and intelligent automation.
- Designed secure REST APIs with JWT, AES encryption, and API gateway policies, ensuring compliance and data protection.
- Optimized MySQL and PostgreSQL with indexing, query refactoring, and schema design, achieving 35% faster query execution.
- Streamlined CI/CD workflows using Jenkins and GitHub Actions, cutting deployment time from weeks to hours.
- Contributed to system architecture discussions, performance tuning, and cloud migration planning.

EDUCATION

Master of Science in Computer Science – The University of Texas at Dallas, TX, USA	May 2024
Bachelor of Technology in Computer Science – National Institute of Technology Patna, India	May 2022

PROJECTS

Scalable Cloud Log Analytics Platform (Distributed Systems + AWS)

- Technologies:** Python, FastAPI, AWS Lambda, DynamoDB, S3, Redis, Docker, CI/CD
- Designed and implemented a distributed log ingestion and analytics platform to process millions of events per day.
 - Built an event-driven pipeline for real-time log ingestion, transformation, and indexing, enabling near real-time search and aggregation.
 - Implemented horizontal scaling, caching strategies, and partitioned data models, reducing query latency by 45% under high concurrency.
 - Designed REST APIs with rate limiting, JWT authentication, and RBAC to secure multi-tenant data access.
 - Added automated deployment using Docker and CI/CD pipelines, ensuring zero-downtime releases.

Cloud Cost Optimization Engine (AWS + Automation)

- Technologies:** AWS (EC2, Lambda, ECS, DynamoDB), Python, REST APIs, React, Serverless Workflows
- Built a cloud resource optimization service analyzing AWS usage and identifying underutilized resources.
 - Developed optimization rules for instance right-sizing, idle resource detection, and storage tiering, achieving simulated cost savings of 25–35%.
 - Exposed recommendations via secure REST APIs (OAuth 2.0, JWT) and visualized metrics through a React-based dashboard.
 - Implemented scheduled jobs and alerting using serverless workflows, enabling proactive cost governance.

Real-Time ML Inference Platform

- Technologies:** TensorFlow, FastAPI, AWS, Python, REST APIs
- Designed a real-time ML inference system enabling low-latency predictions.
 - Built data preprocessing pipelines and feature engineering workflows for model input standardization.
 - Implemented model versioning, A/B testing, and performance monitoring, improving prediction accuracy and deployment reliability.
 - Optimized inference latency with batching, caching, and asynchronous processing, reducing response time by 40%.

Distributed URL Shortening Service (High Availability System)

- Technologies:** Java, Spring Boot, Redis, DynamoDB, AWS ECS, Docker
- Designed and implemented a highly available URL shortening service supporting millions of URLs.
 - Implemented consistent hashing and sharded storage with minimal lookup latency.
 - Built fault-tolerant APIs with rate limiting, caching, and JWT-based authentication.
 - Deployed containerized microservices on AWS ECS with rolling updates and seamless scaling.

Task Management Platform (MERN Stack)

- Technologies:** MongoDB, Express, React, Node.js, JWT
- Designed a scalable full-stack system with JWT-based authentication.
 - Optimized React state management and MongoDB indexing to improve performance and scalability.

Software Maintenance & Enhancement – University of Texas at Dallas

- Technologies:** Java, Backend Systems, QA, Regression Testing
- Refactored legacy components and resolved critical defects across authentication, media synchronization, and data validation.
 - Improved UX with dynamic scrolling, cursor precision, and optimized file retrieval.
 - Applied impact analysis, regression testing, and refactoring, increasing user satisfaction by 12% and reducing downtime.

ACHIEVEMENTS

- LeetCode:** Top 0.4% globally (Dec 2024)
- NPTEL Python Certification:** Top 2% (Apr 2021)